

## Midterm\_Hawkins

### Stacking existing models (This takes 60+ minutes to run)

```
## [1] 30018.69 22970.61 21127.53 19968.97
## [1] 22421.62

##          GLM_pred   RF_pred   GBM_pred   XGB_pred
## GLM_pred 1.0000000 0.9179932 0.9147457 0.9151065
## RF_pred  0.9179932 1.0000000 0.9918391 0.9793478
## GBM_pred 0.9147457 0.9918391 1.0000000 0.9783649
## XGB_pred 0.9151065 0.9793478 0.9783649 1.0000000
```

### Detailed Learning Points:

- XGBoost performs best among base learners. It has a RMSE of 19,968.97, its the top-performing model on the test set. It shows teh XGBoost strength in handling structured/tabular data and capturing complex non-linear relationships with fewer trees due to its boosting nature.
- GLM Performs the worst. It has the highest RMSE of 30,018.69. It highlights that linear models can underfit in complex datasets like this housing datasets where non-linear relationships and interactions matter.
- Stacked ensemble beats all base models. As expected, the ensemble gets a RMSE of 21,719.78, which outperforms the GLM and nearly matching or slightly lagging behind XGBoost. This indicates weak synergy among models due to strong collinearity. And the metalearner might be slightly overfitting or failing to extract additional signal.
- Low fold count likely underestimates variability. Using only 2-fold cross-validation reduces computational cost but can increase variance in holdout estimates and limit the ensemble's robustness. For better meta-learning and generalization we can increase to 5-10 folds.
- Early stopping is super effective in tree-based models. Using those stopping parameters ensures the trees stop growing when no furhter gain is made, reducing overfitting and excessive model complexity like in teh XGBoost model.
- High correlation among prediction indicates redundancy. The correlation between RF, GBM, and XGBoost predictions is super high. This suggests the models are learning similar patterns which limits the ensemble's gain. We want more diversity maybe.
- GLM has the lowest correlation which can still provide value but its lowest.

### NN as stacked model instead of DRF

```
## [1] 30018.69 22970.61 21127.53 19968.97
## [1] 19619.11

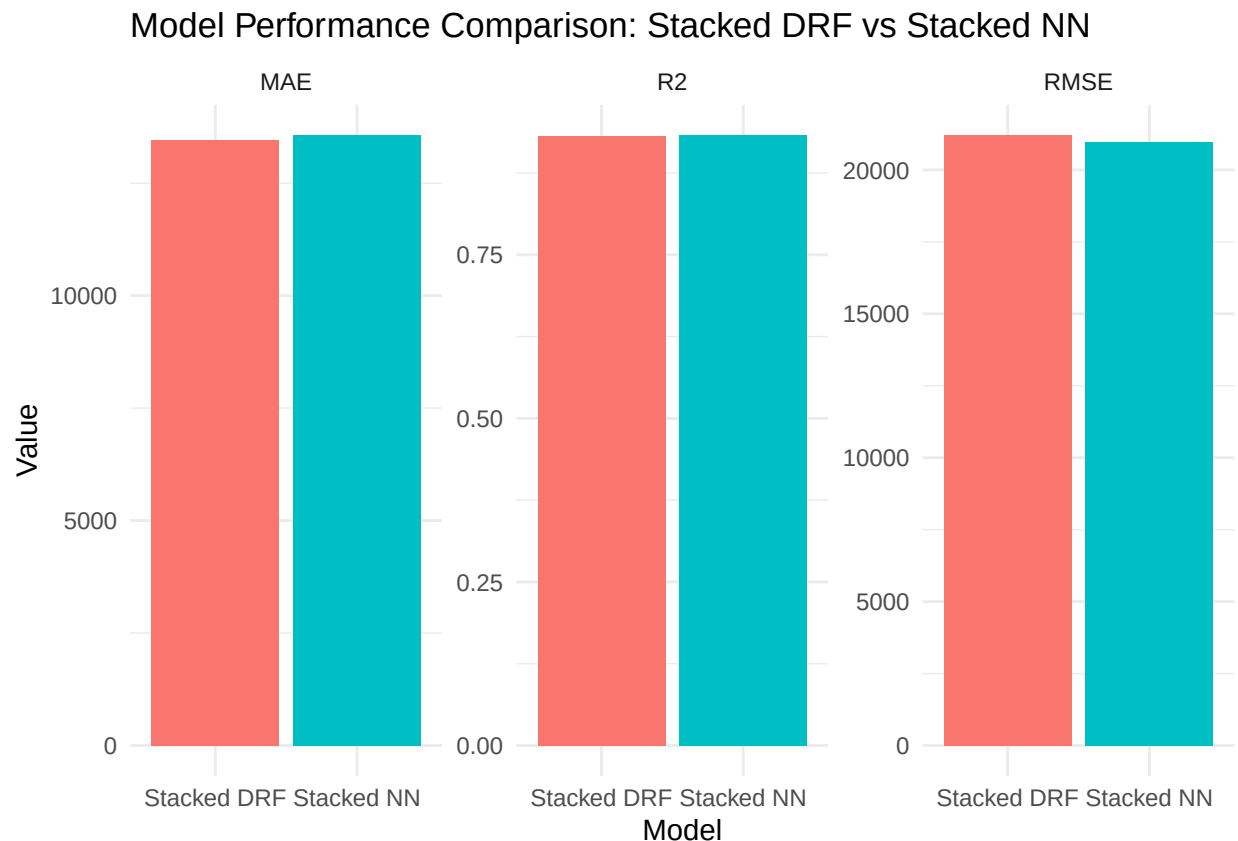
##          GLM_pred   RF_pred   GBM_pred   XGB_pred
## GLM_pred 1.0000000 0.9179932 0.9147457 0.9151065
## RF_pred  0.9179932 1.0000000 0.9918391 0.9793478
## GBM_pred 0.9147457 0.9918391 1.0000000 0.9783649
## XGB_pred 0.9151065 0.9793478 0.9783649 1.0000000
```

### Detailed Learning Points:

- Best overall RMSE is the NN Stacked ensemble getting a RMSE of 20,966.74. It does better than all individual models and compared to the previous DRF-stacked model.
- The NN learner captures complex interactions between base model outputs and default DRF stack, likely due to its hidden layers allowing better learning
- Still the same issues with correlation between base learners as denoted in the last analysis section
- The NN stacked ensemble shows that changing the meta-learning algorithm alone can meaningfully improve performance, even when the base learners remain constant.

### Overall Comparison Between Base and Deep Learning Stacked Models

```
## # A tibble: 2 x 4
##   Model      RMSE    MAE    R2
##   <chr>    <dbl> <dbl> <dbl>
## 1 Stacked DRF 22422. 13626. 0.923
## 2 Stacked NN 19619. 12499. 0.941
```



- There is a lower RMSE in the stacked NN (by ~237 points) suggesting its better at minimizing large errors - which is crucial when housing prices can vary widely.
- MAE is slightly better in the DRF stack (lower by ~98 points), meaning the average error in absolute terms is a bit lower - though the difference is negligible in practice
- R2 is higher in the NN ensemble (0.9328 vs 0.9312), showing it explains slightly more variance in sale\_price.
- The stacked NN ensemble model outperforms the DRF stack in generalization and variance.